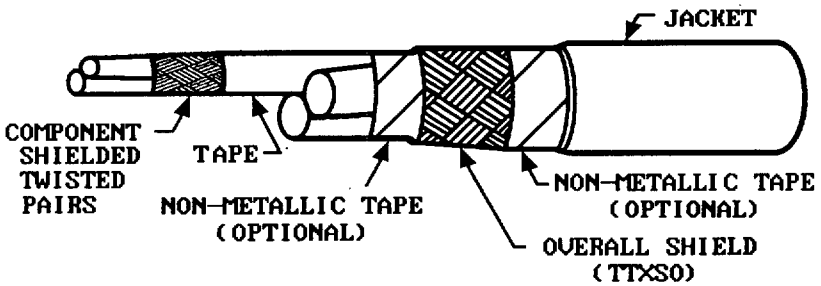
INCH-POUND
MIL-C-24640/5B
31 July 1989
SUPERSEDING
MIL-C-0024640/5A(SH)
23 October 1985
USED IN LIEU OF
MIL-C-24640/5
10 August 1984

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 600 VOLTS, LIGHTWEIGHT,
MULTI-PAIR AWG 20, TYPE TTXS
(INCLUDING VARIATION TYPES TTXSA AND TTXSO)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation: MIL-C-24640.



SH 12751

COMPONENT SHIELDED PAIR:

(B) One black MIL-W-81044/12-20 wire twisted with one white, MIL-W-81044/12-20 wire except identification codes and methods shall be in accordance with MIL-C-24640. Twisted pair identification code applied by method 2 over shields and method 6 over component wires. The shield shall be tin-coated copper braid with a coverage not less than 90 percent determined in accordance with MIL-C-27500.

FIGURE 1. Construction (non-watertight).

(B) denotes changes.

AMSC N/A FSC 6145
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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(B) TABLE I. Construction details-unarmored.

PIN M24640/5	Type number	Number of pairs	Jacket thickness (inch) (minimum)	Cable diameter (inch)		Cable weight (pounds per 1000 feet) (max)
				(min)	(max)	
-01UN	TTXS-2	2	0.038	0.335	0.361	74.3
-02UN	TTXS-4	4	.041	.393	.424	115.0

(B) TABLE II. Construction details-armored.

PIN M24640/5	Type number	Number of pairs	Jacket thickness (inch) (minimum)	Cable diameter (inch)		Cable weight (pounds per 1000 feet) (max)
				(min)	(max)	
-01AN	TTXSA-2	2	0.038	0.385	0.411	100.2
-02AN	TTXSA-4	4	.041	.443	.474	150.4

(B) TABLE III. Construction details-overall shield.

PIN M24640/5	Type number	Number of pairs	Diameter over shield (inch) (nom)	Jacket thickness (inch) (minimum)	Cable diameter (inch)		Cable weight (pounds per 1000 feet) (max)
					(min)	(max)	
-01UO	TTXSO-2	2	0.283	0.041	0.386	0.416	126.0
-02UO	TTXSO-6	6	.418	.044	.506	.546	216.0
-03UO	TTXSO-8	8	.510	.044	.587	.633	284.0
-04UO	TTXSO-10	10	.553	.044	.627	.675	311.0

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REQUIREMENTS:

Qualification required.
Construction details: See tables I, II and III.

When this cable is manufactured with armor, the following additional requirements shall apply:

(B) (a) Braided metal armor shall be added.
(b) Cable surface marking will not be required.

When this cable is manufactured with an overall shield, the following additional requirements shall apply:

(B) (a) An overall tin-coated copper braid shall be added.
(b) Braid coverage shall be not less than 90 percent determined in accordance with MIL-C-27500.

EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
(B) Conductor resistance - ohms/1000 feet, at 25°C, maximum	10.95
(B) Voltage withstand - volts, root mean square, minimum	
Conductor to conductor	2000
Conductor to shield	1000
Shield to shield (TTXSO only)	500
Insulation resistance - megohms/1000 feet, minimum	5000
Conductor and shield continuity (TTXSO only)	No failure
Jacket flaws (TTXSO only)	No failure
Group A:	
Visual and dimensional	No failure
Capacitance:	
Mutual capacitance - at 1 MHz, picofarads/foot (pF/ft), maximum	50
Capacitance unbalance - percent, maximum	15
(B) Characteristic impedance - at 1 MHz, ohms	30 min, 63 max
Watertightness (see table XI of MIL-C-24640 for limits of water leakage)	Not applicable
Weight	No failure

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EXAMINATION AND TESTS: - Continued

	Requirements
Group B:	
Cold bending	No failure
Drip	Zero
Tear - pounds per inch thickness, minimum	35
Physicals (unaged)	
Jacket	
Tensile strength - lb/in², minimum	1300
Elongation - percent, minimum	160
(B) Attenuation - at 3 MHz, dB/100 feet, maximum	3
Gas flame (1 hour)	Not applicable
Group C:	
Physicals (aged)	
Jacket	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged, minimum	60
Hot oil immersion	
Tensile strength - lb/in², minimum	50
Elongation - percent, minimum	50
Permanence of printing (conductor) - cycles, minimum	50
Permanence of printing (jacket) - cycles, minimum	250
Armor (type TTXSA only) - conformance to material, construction and coverage	No failure
Shield - conformance to material, construction and coverage	No failure
Heat distortion - percent unaged, maximum	30
Cable filler removability	No failure
Shrinkage	No failure
Mutual inductance - dB/1000 feet, maximum	--
Surface transfer impedance (type TTXSO only)	
Milliohms per meter, maximum	700
EMP response, dB, minimum	60
Pulse response time - ms/500 feet	
Rise time, maximum	--
Decay time, maximum	--

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QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, and C, plus the following:

	Requirements
Aging and compatibility (cable)	No failure
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Flame propagation	No failure
Immersion	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
Fillers	--
Insulation	10
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer - (type A) hardness, minimum	80
Weathering	No failure
Fungus resistance	No failure
Abrasion resistance	No failure
Gas flame (3 hour)	Not applicable
Finished wire	Not applicable

Preparing activity:
Navy - SH
(Project 6145-1004-05)

Custodians:
Army - MI
Navy - SH
Air Force - 85

Review activities:
Navy - EC
Air Force - 99
DLA - IS

User activities:
Army - ME, AR
Navy - CG

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